

Yes — after comparing:

1. Your **existing spreadsheet tasks** (image),
2. The **latest MoMs/scope clarification**, and
3. The **updated behavioral-process-intelligence direction**,

your existing D6 task structure is actually quite strong technically.
You are NOT far away from the correct scope.

The main issue is not that you added “wrong” tasks.
The issue is mostly:

- terminology,
- removing old “digital twin/optimization” language,
- and explicitly aligning tasks with:
 - behavioral intelligence,
 - expected-vs-actual analysis,
 - anomaly interpretation,
 - HMI visualization support.

Main Comparison Summary

Area	Existing Sheet	Latest Scope Alignment	Action
System IO mapping	Good	Fully aligned	KEEP
KPI categorization	Good	Fully aligned	KEEP
Disturbance modeling	Good	Important	KEEP
Process-state boundaries	Good	Important	KEEP
Proxy simulation/model env	Good	Still useful	KEEP
ARX / state-space SI	Good	Core requirement	KEEP
ML/LSTM wording	Slightly broad	Needs refinement	MODIFY
Training workflows	Good	Fully aligned	KEEP
Residual analysis	Very important	Fully aligned	KEEP

Offline validation	Correct	Matches MoM	KEEP
DT integration wording	Outdated	Misaligned	REMOVE/MODIFY
Expected-vs-actual framework	Missing explicitly	Important	ADD
Expected process envelope	Missing explicitly	Important	ADD
Process-state-aware modeling	Implicit only	Important	ADD
Runtime learning/adaptive AI	Not explicit	Good	DO NOT ADD

UPDATED D6 TASK LIST (Recommended Final Version)

Below is the refined version aligned with:

- latest MoMs,
 - revised scope,
 - process intelligence direction,
 - behavioral process model approach.
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D6 — Ver1 System Identification using Collected Sample Data

(Data-Driven Behavioral Model)

D6.1 — System IO Mapping & KPI Categorization

KEEP MOSTLY AS-IS

Recommended Final Version

- Define process input variables
(temperature, pressure, laying speed, tension, TCP position, machine states)
 - Define target outputs
(process deviation, KPI trends, stability indicators, defect indicators, process-health indicators)
 - Categorize KPIs into:
 - quality KPIs,
 - process KPIs,
 - machine KPIs
 - Define disturbance variables and external influences
 - Define behavioral boundaries between process states
(layup, retract, cut/add, transition states)
 - Define expected process behavior and nominal operating envelopes
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Why this modification?

Added:

expected process behavior
nominal operating envelopes

because this became central in the new scope.

D6.2 — Proxy Plant / Behavioral Modeling Environment Setup

Mostly KEEP

Recommended Final Version

- Setup Python/MATLAB behavioral modeling environment
 - Define reusable behavioral modeling workflows
 - Develop mock process behavior generators using sample datasets
 - Create baseline process-response simulation environment
 - Create expected-vs-actual comparison workflow for model evaluation
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Why?

The old wording:

simulation environment

could sound like:

- full digital twin,
- virtual commissioning.

Now it is framed as:

process-response evaluation environment

which is safer.

D6.3 — Implement Behavioral Modeling / System Identification Algorithms

MODIFY slightly

Recommended Final Version

- Implement baseline SI methods
(ARX, non-ARX, state-space, regression-based approaches)
- Evaluate temporal behavioral modeling approaches
- Evaluate hybrid or physics-informed behavioral approaches
- Define model training workflows

- Define parameter estimation workflows
 - Develop process-state-aware behavioral modeling
 - Compare model complexity vs prediction accuracy
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REMOVE THIS OLD LINE

Remove:

Evaluate ML/LSTM or time series models based approaches

because:

- time-series anomaly monitoring is D5,
 - D6 is behavioral modeling/SI.
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Replace with:

Evaluate temporal behavioral modeling approaches

Much cleaner.

D6.4 — Initial Behavioral Model Training

Mostly KEEP

Recommended Final Version

- Train SI/behavioral models using prepared datasets
- Calibrate model parameters
- Tune model hyperparameters
- Improve prediction consistency and operational stability
- Validate transient and steady-state process behavior representation
- Generate expected process envelopes and nominal KPI operating regions
- Compare predicted vs expected process behavior
- Generate expected KPI evolution trends

Why?

Added:

expected process envelopes
expected KPI evolution

These are now core to:

- anomaly analysis,
 - HMI,
 - expected-vs-actual logic.
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D6.5 — Offline Validation Loop 1

KEEP with one addition

Recommended Final Version

- Perform train/validation/test evaluation
 - Compute prediction accuracy metrics
 - Perform residual/error analysis
 - Evaluate process-behavior prediction quality
 - Validate model robustness across operational scenarios
 - Evaluate model generalization capability
 - Generate expected-vs-actual deviation indicators for anomaly interpretation and visualization
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Why add this?

This is now central to:

- anomaly detection,
- process intelligence,

- HMI integration.
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D6.6 — Residual Analysis & Model Refinement

Slight refinement

Recommended Final Version

- Analyze prediction residuals
 - Identify unstable operating regions
 - Refine feature selection
 - Improve behavioral model structure
 - Retune parameters and retrain if required
 - Improve model consistency and operational stability
 - Refine expected process envelopes and KPI boundaries
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Why?

Changed:

Improve prediction robustness

to:

Improve operational stability

This matches the latest scope better.

D6.7 — Initial Behavioral Model Release

MODIFY IMPORTANTLY

Recommended Final Version

- Freeze validated Ver1 behavioral model
 - Document model assumptions and limitations
 - Prepare interfaces for:
 - anomaly-analysis framework
 - expected-vs-actual comparison
 - HMI visualization integration
 - Document operational KPI envelopes and behavioral assumptions
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IMPORTANT REMOVE

Remove:

Prepare model interfaces for DT integration

This no longer aligns well with the revised scope.

MOST IMPORTANT OBSERVATION

Your original spreadsheet was already technically good.

The biggest scope correction is:

OLD Direction

Digital Twin
Optimization Engine
Runtime Adaptive AI
Virtual Commissioning

NEW Direction

Behavioral Process Intelligence
Expected Process Behavior
Process Variation Understanding
Anomaly Interpretation
Expected-vs-Actual Comparison
HMI Visualization

And your updated D6 after these refinements aligns very well with the new direction.